

Simulating Just-in-Time Adaptive Interventions: A Config-Driven, LLM-Bootstrapped Framework

William Dennis

University of Bristol & National University of Singapore

Physical inactivity is a leading modifiable health risk [1]; just-in-time adaptive interventions (JITAI) deliver personalised prompts through wearables, but trials cannot probe counterfactuals and transition data is rarely public [2]. We reduce policy design to a config-driven pipeline for two simulators: LLM-bootstrapped patient models, evaluated against an optimal dynamic-programming bound and bandit baselines.

1. Motivation and approach

We formalise the intervention as a Markov decision process $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R})$ and build a framework in which *everything* is configuration: state spaces, action sets, transition models, rewards, and agents are declared in YAML, so experiments differ by config file alone (configs and tables in the repository) [3, 4].

2. LLM-bootstrapped transitions

The transition function $P(s' | s, a)$ is the hardest component of any patient model: it must encode how context, burden, and a delivered suggestion change behaviour, and it is rarely available as data. Rather than hand-curating tables or waiting for trials, we prompt an LLM (DeepSeek V4 Flash, with a GLM 5.2 comparison) to sample next-state distributions given a persona description, current state, and candidate action — a text world model [5]. Unlike data-driven simulators [2], which require logged trial data, bootstrapping needs no logged data — though it still requires hand-authored personas, states, and rewards in config. Personas were hand-curated by domain practitioners, loosely grounded in the behaviour-change literature: not arbitrary, but not validated against real patient data either. Park et al. [6] report 85% accuracy on General Social Survey replication for their agents; we make no such population-replication claim for ours. Six sweeps of 22,320 prompts each (133,920 prompts, 10 samples per outcome) produced 31 JSON tables: \$24.52 (DeepSeek) + \$25.45 (GLM 5.2). Personas produced distinct sampled transition distributions, and the config’s Bayesian mechanism models burden as lowering future activity probability [2, 7].

3. Validation

On the Phase-1 persona tables, a HeartSteps-V2-style UCB (upper confidence bound) bandit reaches 47–95.8% of the optimal dynamic-programming bound across personas (50 seeds; 95% confidence intervals ± 0.6 –9.5pp); the idle policy alone reaches 96.8–100.1%, the values above 100% being Monte Carlo noise (± 3 –5) — the bound is essentially the idle policy — interventions underperform idle on these tables, a property of these personas’ bootstrapped tables, not a claim about JITAI efficacy in general. All baselines are internal to the bootstrapped tables (mock-on-mock; no ground truth; burden sensitivity untested). Eight literature-review reports document this background; the one quantitative comparison (a V2-style agent at 46.7% vs the paper’s 78.4%, on different data) did not reproduce.

4. PEARL-matched extension (Phase 2)

In the final two weeks we compared against the PEARL randomised controlled trial (RCT) [8], whose reinforcement-learning (RL) arm gained +295.9 steps over control ($p=0.0002$): we used 6 COM-B themes $\times$ 2 delivery times [9], plus idle — with an ϵ -greedy bandit and a COM-B barrier-weighted fixed arm [10–12]. The 14,040-prompt bootstrap (108 states $\times$ 13 actions $\times$ 10 samples) passed 11 of 17 checks; the six failures (inflated effect size $d=1.525$; fixed $>$ RL inversion; no attenuation; no burden saturation; inverted weekend; degenerate control t-test) trace to an idle-vs-action structure. A reward-penalty probe ($R = \text{step_reward} - \lambda \mathbf{1}[\text{action} \neq \text{idle}]$, $\lambda = 0.05$, $\text{step_reward} -1/0/+1$; an anti-spam term from the Phase-1 reward design, independent of the PEARL comparison) does not restore the RL-vs-fixed ordering: 50-seed runs sit within noise (± 5 –7) and the ordering flips with the penalty (unpenalised RL 3.0 $>$ fixed 2.4; penalised fixed $-0.6 >$ RL -0.7) — the penalty taxes activeness. Calibration metrics, not trial outcomes; the probe is not a validated fix.

5. Conclusion

Policy design reduces to a cheap, config-driven pipeline, albeit mock-on-mock only, with a documented map of where the simulation diverges from published trials.

References

- [1] R. Guthold, G. A. Stevens, L. M. Riley, and F. C. Bull. Worldwide trends in insufficient physical activity from 2001 to 2016: a pooled analysis of 358 population-based surveys with 1.9 million participants. *The Lancet Global Health*, 6(10):e1077–e1086, 2018. doi: 10.1016/S2214-109X(18)30357-7.
- [2] S. Wang, C. Zhang, B. Kröse, and H. van Hoof. Optimizing adaptive notifications in mobile health interventions: Reinforcement learning from a data-driven behavioral simulator. *Journal of Medical Systems*, 45(12):102, 2021. doi: 10.1007/s10916-021-01773-0.
- [3] P. Klasnja, S. Smith, N. J. Seewald, A. Lee, et al. Efficacy of contextually tailored suggestions for physical activity: A micro-randomized trial. *Annals of Behavioral Medicine*, 53(6):573–583, 2019.
- [4] A. L. Trella et al. Designing reinforcement learning for digital interventions. *Algorithms*, 15(8):255, 2022. doi: 10.3390/a15080255.
- [5] B. Li and W. Y. Wang. Bridging the agent-world gap: Text world models for llm-based agents. *arXiv preprint*, 2026. arXiv:2606.09032.
- [6] J. S. Park, C. Q. Zou, A. Shaw, B. M. Hill, C. J. Cai, M. R. Morris, R. Willer, P. Liang, and M. S. Bernstein. Generative agent simulations of 1,000 people. *arXiv preprint*, 2024. arXiv:2411.10109.
- [7] J. S. Park, J. C. O’Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proc. ACM UIST*, 2023.
- [8] A. A. Lee, N. Hegde, N. Deliu, E. Rosenzweig, A. Suggala, S. Lakshminarasimhan, Q. He, J. Hernandez, M. Seneviratne, R. Singh, P. Kalkar, K. Shanmugam, A. Raghuveer, A. Singh, M. Nguyen, J. Taylor, J. Alla, S. S. Villar, and H. Emir-Farinas. A personalized exercise assistant using reinforcement learning (PEARL): Results from a four-arm randomized-controlled trial, January 2026. JMIR preprint (doi: 10.2196/preprints.91156); pre-registered on OSF (doi: 10.17605/OSF.IO/TW7UP).
- [9] S. Michie, M. M. van Stralen, and R. West. The behaviour change wheel: A new method for characterising and designing behaviour change interventions. *Implementation Science*, 6(1):42, 2011. doi: 10.1186/1748-5908-6-42.
- [10] X. Liu, N. Deliu, T. Chakraborty, L. Bell, and B. Chakraborty. Thompson sampling for zero-inflated count outcomes with an application to the Drink Less mobile health study. *The Annals of Applied Statistics*, 19(2):1403–1425, 2025. doi: 10.1214/25-AOAS2030.
- [11] S. Ghosh, Y. Guo, P.-Y. Hung, L. Coughlin, E. Bonar, I. Nahum-Shani, M. Walton, and S. Murphy. reBandit: Random effects based online RL algorithm for reducing cannabis use. In *Proc. IJCAI*, pages 7278–7286, 2024. doi: 10.24963/ijcai.2024/805.
- [12] S. Tomkins, P. Liao, P. Klasnja, and S. Murphy. IntelligentPooling: practical Thompson sampling for mHealth. *Machine Learning*, 110:2685–2727, 2021. doi: 10.1007/s10994-021-05995-8.